

12.263

EE25BTECH11042 - Nipun Dasari

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Question

The matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{pmatrix} \quad (0.1)$$

can be decomposed into the product of a lower triangular matrix $\mathbf{L}$ and an upper triangular matrix $\mathbf{U}$ as $\mathbf{A} = \mathbf{LU}$. Let $\mathbf{x}, \mathbf{z} \in \mathbb{R}^3$ and $\mathbf{b} = (1 \ 1 \ 1)^\top$.

a) The solution $\mathbf{z} = (z_1 \ z_2 \ z_3)^\top$ of the system $\mathbf{Lz} = \mathbf{b}$ is

- i) $(-1 \ -1 \ -2)^\top$
- ii) $(1 \ -1 \ 2)^\top$
- iii) $(1 \ -1 \ -2)^\top$
- iv) $(-1 \ 1 \ 2)^\top$

b) The solution $\mathbf{x} = (x_1 \ x_2 \ x_3)^\top$ of the system $\mathbf{Ux} = \mathbf{z}$ is

- i) $(2 \ 1 \ -2)^\top$
- ii) $(2 \ 1 \ 2)^\top$
- iii) $(-2 \ -1 \ -2)^\top$
- iv) $(-2 \ 1 \ -2)^\top$

Theoretical solution

We will perform row operations on $\mathbf{A}$ to obtain $\mathbf{U}$. The multipliers used will form the entries of $\mathbf{L}$.

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 2 \end{pmatrix} \quad (0.2)$$

Operation $R_2 \rightarrow R_2 - (\textcolor{red}{2})R_1$. The multiplier is $l_{21} = 2$.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 1 & 3 & 2 \end{pmatrix} \quad (0.3)$$

Operation $R_3 \rightarrow R_3 - (\textcolor{red}{1})R_1$. The multiplier is $l_{31} = 1$.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 2 & 1 \end{pmatrix} \quad (0.4)$$

Theoretical solution

Operation $R_3 \rightarrow R_3 - (-2)R_2$. The multiplier is $l_{32} = -2$.

$$\mathbf{U} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (0.5)$$

The lower triangular matrix $\mathbf{L}$ is constructed from the multipliers.

$$\mathbf{L} = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -2 & 1 \end{pmatrix} \quad (0.6)$$

Theoretical solution

Part a): Solving $\mathbf{Lz} = \mathbf{b}$

We solve the system using Gaussian elimination.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & -2 & 1 & 1 \end{array} \right] \xleftrightarrow[\begin{array}{c} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}]{\quad} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & -2 & 1 & 0 \end{array} \right] \quad (0.7)$$

$$\xleftrightarrow{R_3 \rightarrow R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 \end{array} \right] \quad (0.8)$$

The solution is $\mathbf{z} = (1 \quad -1 \quad -2)^\top$, which is option (iii).

Theoretical solution

Part b): Solving $\mathbf{U}\mathbf{x} = \mathbf{z}$

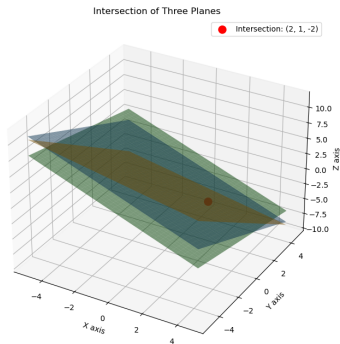
We solve the system using the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & -2 \end{array} \right] \xleftrightarrow{R_2 \rightarrow -R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array} \right] \quad (0.9)$$

$$\xleftrightarrow[\begin{array}{l} R_1 \rightarrow R_1 - R_3 \\ R_1 \rightarrow R_1 - R_2 \end{array}]{\begin{array}{l} R_1 \rightarrow R_1 - R_3 \\ R_1 \rightarrow R_1 - R_2 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array} \right] \quad (0.10)$$

The solution is $\mathbf{x} = (2 \ 1 \ -2)^\top$, which is option (i).

Plot



Fig